

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12404816
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Hamamoto; Ryo et al.

Diagnostic device for exhaust gas aftertreatment device, diagnostic method for exhaust gas aftertreatment device, and diagnostic program for exhaust gas aftertreatment device

Abstract

A diagnostic device for an exhaust gas aftertreatment device includes a temperature detection device that detects a filter inlet-side temperature and a filter outlet-side temperature, a differential pressure detection device that detects a differential pressure between a filter inlet-side pressure and a filter outlet-side pressure, an intake air pressure detection device that detects an intake air pressure of an engine, and a controller that determines a malfunction in an exhaust gas aftertreatment device based on values detected by the temperature detection device, the differential pressure detection device, and the intake air pressure detection device.

Inventors:	Hamamoto; Ryo (Hiroshima, JP), Nakagawa; Tomohiro (Hiroshima, JP), Endo; Kazuomi (Hiroshima, JP), Togo; Hiroshi (Hiroshima, JP)
Applicant:	KOBELCO CONSTRUCTION MACHINERY CO., LTD (Hiroshima, JP)
Family ID:	87552225
Assignee:	KOBELCO CONSTRUCTION MACHINERY CO., LTD (Hiroshima, JP)
Appl. No.:	18/833443
Filed (or PCT Filed):	January 24, 2023
PCT No.:	PCT/JP2023/002124
PCT Pub. No.:	WO2023/149285
PCT Pub. Date:	August 10, 2023

Prior Publication Data

Foreign Application Priority Data

JP 2022-014996 Feb. 02, 2022

Publication Classification

Int. Cl.: F02D35/00 (20060101); F01N3/021 (20060101); F01N11/00 (20060101)

U.S. Cl.:

CPC F02D35/0015 (20130101); F01N3/021 (20130101); F01N11/002 (20130101);

Field of Classification Search

CPC: F01N (3/00); F01N (3/021); F01N (9/002); F02D (45/00); F02D (35/0015); G06F (21/64)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9074505	12/2014	Sase	N/A	F01N 9/002
9151015	12/2014	Noguchi	N/A	F02D 41/222
10125652	12/2017	Kurisaka	N/A	F02B 37/007
11236654	12/2021	Noguchi	N/A	F01N 3/10
2007/0251214	12/2006	Nishino	N/A	N/A
2014/0238003	12/2013	Sase	60/311	F02D 41/1467
2014/0308170	12/2013	Takayanagi	422/119	F02D 41/029
2017/0051651	12/2016	Kurisaka	N/A	F01N 3/0814
2017/0250961	12/2016	Rasbornig	N/A	N/A
2019/0218753	12/2018	Morimoto	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2 037 090	12/2008	EP	N/A
2 199 552	12/2009	EP	N/A
2007-292013	12/2006	JP	N/A
2015-148182	12/2014	JP	N/A
2016-70077	12/2015	JP	N/A
2016-79811	12/2015	JP	N/A
2017-48709	12/2016	JP	N/A
2017-153082	12/2016	JP	N/A
WO 2018/062363	12/2004	WO	N/A

OTHER PUBLICATIONS

Machine translation, JP2016079811, Katsuta et al., Specification, obtained from <https://worldwide.espacenet.com/>, originally published May 16, 2016, pp. 1-7. cited by examiner

Machine translation, JP2015148182, Noguchi et al., Specification, obtained from <https://worldwide.espacenet.com/>, originally published Aug. 20, 2015, pp. 1-15. cited by examiner
International Search Report Issued Mar. 7, 2023 in PCT/JP2023/002124, filed on Jan. 24, 2023, (2 pages). cited by applicant
Extended European Search Report issued Oct. 23, 2024 in European Patent Application No. 23749601.3, 8 pages. cited by applicant

Primary Examiner: Moubry; Grant

Attorney, Agent or Firm: Oblon, McClelland, Maier & Neustadt, L.L.P.

Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a diagnostic device for an exhaust gas aftertreatment device, a diagnostic method for an exhaust gas aftertreatment device, and a diagnostic program for an exhaust gas aftertreatment device, the diagnostic device, the diagnostic method, and the diagnostic program diagnosing a malfunction in an exhaust gas aftertreatment device provided in a vehicle.

BACKGROUND ART

(2) Patent Literature 1 discloses a particulate filter malfunction diagnostic device that determines a malfunction in a particulate filter based on a deposition amount of particulate matter (PM). A PM sensor detects the deposition amount of PM based on a change in electric resistance between a pair of electrodes.

(3) In addition, Patent Literature 2 discloses a diagnostic device that includes a PM sensor including at least a pair of electrode members forming a capacitor, the PM sensor being provided in a filter member disposed downstream of a diesel particulate filter (DPF), and that determines a malfunction in the DPF based on electrostatic capacitance between the electrode members.

(4) However, with the PM sensor of Patent Literature 1, a great change in an output is inevitable due to adhesion of PM having a large particle size to a portion between the pair of electrode members. Further, in Patent Literature 2, a capacitor has to be added, and an improvement in DPF is necessary.

CITATION LIST

Patent Literature

(5) Patent Literature 1: JP 2017-48709 A Patent Literature 2: JP 2016-70077 A

SUMMARY OF INVENTION

(6) An object of the present invention is to provide a diagnostic device for an exhaust gas aftertreatment device, a diagnostic method for the exhaust gas aftertreatment device, and a diagnostic program for the exhaust gas aftertreatment device, the diagnostic device, diagnostic method, and diagnostic program being capable of detecting a malfunction in the exhaust gas aftertreatment device with high accuracy.

(7) Provided is a diagnostic device for an exhaust gas aftertreatment device that has a filter for collecting fine particles from exhaust gas of an engine and is provided in a vehicle, the diagnostic device diagnosing a malfunction in the exhaust gas aftertreatment device, the diagnostic device including: a temperature detection device that detects a filter inlet-side temperature and a filter outlet-side temperature; a differential pressure detection device that detects a differential pressure between a filter inlet-side pressure and a filter outlet-side pressure; an intake air pressure detection device that detects an intake air pressure of the engine; and a controller that determines a malfunction in the exhaust gas aftertreatment device based on values detected by the temperature

detection device, the differential pressure detection device, and the intake air pressure detection device.

(8) Further, provided is a diagnostic method for an exhaust gas aftertreatment device that has a filter for collecting fine particles from exhaust gas of an engine and is provided in a vehicle, the diagnostic method for diagnosing a malfunction in the exhaust gas aftertreatment device, the diagnostic method including: a temperature detection step of detecting a filter inlet-side temperature and a filter outlet-side temperature; a differential pressure detection step of detecting a differential pressure between a filter inlet-side pressure and a filter outlet-side pressure; an intake air pressure detection step of detecting an intake air pressure of the engine; and a determination step of determining a malfunction in the exhaust gas aftertreatment device based on values detected in the temperature detection step, the differential pressure detection step, and the intake air pressure detection step.

(9) Further, provided is a diagnostic program for an exhaust gas aftertreatment device that has a filter for collecting fine particles from exhaust gas of an engine and is provided in a vehicle, the diagnostic program for causing a computer to diagnose a malfunction in the exhaust gas aftertreatment device, the diagnostic program including causing the computer to determine a malfunction in the exhaust gas aftertreatment device based on a filter inlet-side temperature, a filter outlet-side temperature, a differential pressure between a filter inlet-side pressure and a filter outlet-side pressure, and an intake air pressure of the engine.

(10) In the present invention, a malfunction in the exhaust gas aftertreatment device is determined based on the filter inlet-side temperature, the filter outlet-side temperature, the differential pressure between the filter inlet-side pressure and the filter outlet-side pressure, and the intake air pressure of the engine. The deposition amount of fine particles deposited on the filter can be obtained from the differential pressure between the filter inlet-side pressure and the filter outlet-side pressure. A temperature state of the filter is obtained from the filter inlet-side temperature and the filter outlet-side temperature. Therefore, before the filter is damaged by melting, a malfunction in the exhaust gas aftertreatment device can be detected by using the differential pressure between the filter inlet-side pressure and the filter outlet-side pressure, the filter inlet-side temperature, and the filter outlet-side temperature for determination. In the present invention, unlike the PM sensor of Patent Literature 1, the outputs from the temperature detection device, the differential pressure detection device, and the intake air pressure detection device do not change significantly. Therefore, the malfunction in the exhaust gas aftertreatment device can be detected with high accuracy.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a side view of a work machine.
- (2) FIG. 2 is a configuration diagram illustrating a hydraulic circuit of the work machine.
- (3) FIG. 3 is a configuration diagram of an exhaust gas aftertreatment device.
- (4) FIG. 4 is a configuration diagram illustrating an electric circuit of the work machine.
- (5) FIG. 5 is a flowchart of diagnostic processing.

DESCRIPTION OF EMBODIMENTS

(6) A preferred embodiment of the present invention will be described below with reference to the drawings.

Configuration of Work Machine

(7) A diagnostic device (diagnostic device **1**) for an exhaust gas aftertreatment device according to an embodiment of the present invention diagnoses a malfunction in the exhaust gas aftertreatment device provided in a vehicle. As illustrated in FIG. 1 which is a side view of a work machine **20**, the diagnostic device **1** is provided in the work machine **20**. The work machine **20** is one example

- of a vehicle, and is, for example, a hydraulic excavator. The work machine **20** may be operated by an operator in a cab **23**, may be remotely operated, or may be automatically operated.
- (8) The work machine **20** has a machine main body **25** including a lower travelling body **21** and an upper slewing body **22**, an attachment **30**, and cylinders **40**.
- (9) The lower travelling body **21** is a portion that causes the work machine **20** to travel, and includes, for example, right and left crawlers **26**. The upper slewing body **22** is slewably attached to an upper part of the lower travelling body **21** via a slewing device **24**. The cab **23** is provided at a front part of the upper slewing body **22**.
- (10) The attachment **30** is attached to the upper slewing body **22** to be rotatable in an up-and-down direction. The attachment **30** includes a boom **31**, an arm **32**, and a bucket **33**. The boom **31** is attached to the upper slewing body **22** to be rotatable in the up-and-down direction (capable of being raised and lowered). The arm **32** is attached to the boom **31** to be rotatable in the up-and-down direction. The bucket **33** is attached to the arm **32** to be rotatable in a front-and-rear direction. The bucket **33** is a tip attachment that is a tip portion of the attachment **30**, and is a part that performs work such as excavation, leveling, and scooping of earth and sand. Note that a work object to be held by the bucket **33** is not limited to earth and sand, but may also be stones or waste (such as industrial waste). The tip attachment is not limited to the bucket **33**, and may be a grapple, a lifting magnet, or the like.
- (11) The cylinders **40** can hydraulically turn the attachment **30**. The cylinders **40** are hydraulic telescopic cylinders. The cylinders **40** include a boom cylinder **41**, an arm cylinder **42**, and a bucket cylinder **43**.

- (12) The boom cylinder **41** rotates the boom **31** with respect to the upper slewing body **22**. The boom cylinder **41** has a base end portion rotatably attached to the upper slewing body **22**. The boom cylinder **41** has a tip portion rotatably attached to the boom **31**.
- (13) The arm cylinder **42** rotates the arm **32** with respect to the boom **31**. The arm cylinder **42** has a base end portion rotatably attached to the boom **31**. The arm cylinder **42** has a tip portion rotatably attached to the arm **32**.
- (14) The bucket cylinder **43** rotates the bucket **33** with respect to the arm **32**. The bucket cylinder **43** has a base end portion rotatably attached to the arm **32**. The bucket cylinder **43** has a tip portion rotatably attached to a link member **34** rotatably attached to the bucket **33**.

Configuration of Hydraulic Circuit of Work Machine

- (15) As illustrated in FIG. 2 which is a configuration diagram illustrating the hydraulic circuit of the work machine **20**, the work machine **20** includes an engine **51**, a hydraulic pump **52**, and a control valve group **53**. The engine **51** is a power source of the work machine **20**. The hydraulic pump **52** is driven by the engine **51**. The hydraulic pump **52** includes a left hydraulic pump **52L** and a right hydraulic pump **52R**. The control valve group **53** changes a flow rate of hydraulic oil supplied from the hydraulic pump **52** to the cylinders **40** and the like. The control valve group **53** includes a left control valve group **53L** and a right control valve group **53R**.
- (16) The left hydraulic pump **52L** supplies hydraulic oil to each of a left traveling motor **54L**, a slewing motor **55**, and the arm cylinder **42**. The right hydraulic pump **52R** supplies hydraulic oil to each of a right traveling motor **54R**, the boom cylinder **41**, and the bucket cylinder **43**. Note that FIG. 2 illustrates one example of the arrangement of these components, and the arrangement of these components is not limited to FIG. 2. The left traveling motor **54L** drives the left crawler **26** of the lower travelling body **21**, and the right traveling motor **54R** drives the right crawler **26** of the lower travelling body **21**. The slewing motor **55** is provided in the slewing device **24** and slews the upper slewing body **22**.
- (17) The left control valve group **53L** includes a control valve that changes the flow rate of the hydraulic oil supplied to the left traveling motor **54L**, a control valve that changes the flow rate of the hydraulic oil supplied to the slewing motor **55**, and a control valve that changes the flow rate of the hydraulic oil supplied to the arm cylinder **42**. The right control valve group **53R** includes a

control valve that changes the flow rate of the hydraulic oil supplied to the right traveling motor **54R**, a control valve that changes the flow rate of the hydraulic oil supplied to the boom cylinder **41**, and a control valve that changes the flow rate of the hydraulic oil supplied to the bucket cylinder **43**. These control valves are controlled by a controller **71** in response to an operation supplied to an operation device **28** (see FIG. 1) including at least one operation lever provided in the cab **23**. Note that in the present embodiment, the operation device **28** includes a plurality of operation levers, and these operation levers are disposed in the cab **23**.

(18) The engine **51**, the hydraulic pump **52**, and the control valve group **53** are controlled by the controller **71** described later.

(19) As illustrated in FIG. 1, the left and right crawlers **26**, the upper slewing body **22**, and the attachment **30** are movable portions of the work machine **20**. The operation device **28** outputs operation commands for operating the movable portions. As illustrated in FIG. 2, the engine **51**, the hydraulic pump **52**, and the control valve group **53** are power sources that supply power to the movable portions.

Configuration of Exhaust Gas Aftertreatment Device

(20) As illustrated in FIG. 3 which is a configuration diagram of the exhaust gas aftertreatment device **10**, the work machine **20** includes the exhaust gas aftertreatment device **10**. The exhaust gas aftertreatment device **10** is provided in an exhaust pipe **15** through which an exhaust gas of the engine **51** passes, and collects fine particles from the exhaust gas. The exhaust gas aftertreatment device **10** includes a filter **11** that collects fine particles, and an oxidation catalyst **12** that is provided downstream of the filter **11** and oxidizes unburnt fuel.

(21) The diagnostic device **1** includes temperature sensors **2** (one example of a temperature detection device), a differential pressure sensor **3** (one example of a differential pressure detection device), and an intake air pressure sensor **4** (one example of an intake air pressure detection device). The temperature sensors **2** respectively detect a filter inlet-side temperature and a filter outlet-side temperature. The differential pressure sensor **3** detects a differential pressure between a filter inlet-side pressure and a filter outlet-side pressure (for example, a differential pressure between an inlet side and an outlet side of the filter **11**). The intake air pressure sensor **4** is provided in an intake air pipe **16** that supplies air to the engine **51**, and detects an intake air pressure (boost pressure) of the engine **51**.

(22) The filter inlet-side temperature is a temperature detected by the temperature sensor **2** disposed at an exhaust gas inlet of the filter **11** or near the inlet. Specifically, the filter inlet-side temperature may be a temperature of the exhaust gas at the inlet of the filter **11** or the temperature of the filter **11** at the inlet of the filter **11**. Further, the filter inlet-side temperature may be a temperature of the exhaust gas near the inlet upstream of the inlet of the filter **11** or a temperature of the pipe near the inlet upstream of the inlet of the filter **11**.

(23) The filter outlet-side temperature is a temperature detected by the temperature sensor **2** disposed at the exhaust gas outlet of the filter **11** or near the outlet. Specifically, the filter outlet-side temperature may be a temperature of the exhaust gas at the outlet of the filter **11** or a temperature of the filter **11** at the outlet of the filter **11**. Further, the filter outlet-side temperature may be a temperature of the exhaust gas near the outlet downstream of the outlet of the filter **11**, or may be a temperature of the pipe near the outlet downstream of the outlet of the filter **11**.

(24) The filter inlet-side pressure may be a pressure at the inlet of the filter **11**, or may be a pressure near the inlet upstream of the inlet of the filter **11**. The filter outlet-side pressure may be a pressure at the outlet of the filter **11**, or may be a pressure near the outlet downstream of the outlet of the filter **11**. Each of “upstream” and “downstream” is based on the flow direction of the exhaust gas.

Configuration of Electric Circuit of Work Machine

(25) As illustrated in FIG. 4 which is a configuration diagram of an electric circuit of the work machine **20**, the work machine **20** includes the controller **71**, a storage device **73**, and a communication device **74**.

(26) The controller **71** includes a computer that inputs and outputs signals, performs arithmetic (processing), stores information, and the like. For example, the controller **71** includes determination means, notification means, limiting means, accumulated time calculation means, and security processing means, described later. Functions of the determination means, the notification means, the limiting means, the accumulated time calculation means, and the security processing means in the controller **71** are implemented by executing programs stored in the storage unit (not illustrated) of the controller **71**. The controller **71** may be provided at only one place or may be provided separately at a plurality of places. The controller **71** may be mounted on the work machine **20** or may be disposed in a server or the like outside the work machine **20**. The diagnostic device **1** includes the controller **71**.

(27) The controller **71** receives the temperature values respectively detected by the temperature sensors **2**. The controller **71** handles, for example, a temperature value expressed as an absolute temperature. Further, the controller **71** receives a differential pressure value detected by the differential pressure sensor **3**. Further, the controller **71** receives a pressure value detected by the intake air pressure sensor **4**. The controller **71** periodically acquires the temperature values respectively detected by the temperature sensors **2**, the differential pressure value detected by the differential pressure sensor **3**, and the pressure value detected by the intake air pressure sensor **4**.

(28) Further, the controller **71** receives an operation command from the operation device **28**.

(29) The communication device **74** can transmit and receive data by performing communication such as wireless communication or wired communication with a server or the like outside the work machine **20**.

(30) The controller **71** (specifically, the determination means) determines a malfunction in the exhaust gas aftertreatment device **10** based on the values detected respectively by the temperature sensors **2**, the differential pressure sensor **3**, and the intake air pressure sensor **4**. Specifically, for example, the controller **71** may determine a malfunction in the exhaust gas aftertreatment device **10** based on an index. The index may be a value based on a ratio between the filter inlet-side temperature and the filter outlet-side temperature and a ratio between the differential pressure (differential pressure between the filter inlet-side pressure and the filter outlet-side pressure) and the intake air pressure of the engine **51**. The controller **71** may determine that the exhaust gas aftertreatment device **10** has malfunctioned when the index exceeds a threshold.

(31) More specifically, in a case where a value obtained by dividing the filter outlet-side temperature by the filter inlet-side temperature is represented by “a”, and a value obtained by dividing the differential pressure between the filter inlet-side pressure and the filter outlet-side pressure by the intake air pressure of the engine **51** is represented by “b”, the index may be a value represented by “a/b”.

(32) In a case where the filter **11** is clogged, the ratio of the differential pressure to the intake air pressure of the engine **51** in the exhaust gas aftertreatment device **10** is greater than the ratio in the normal state, and the value of “b” is greater. The case where the filter **11** is clogged is a state where a large amount of fine particles are deposited on the filter **11**, and burning the fine particles in this state may cause a damage of the filter **11** due to melting. On the other hand, when the value of “b” increases, the index “a/b” falls below a lower limit value (threshold) in the normal state. Therefore, before the filter **11** is damaged by melting, the controller **71** can detect a malfunction in the exhaust gas aftertreatment device **10** by detecting that the index falls below the lower limit value.

(33) The higher the temperature of the filter **11**, the greater the value of “a”, which is the ratio between the filter inlet-side temperature and the filter outlet-side temperature. In a case where the temperature of the filter **11** becomes high, a damage of the filter **11** due to melting might occur. On the other hand, when the value of “a” increases, the index “a/b” exceeds an upper limit value (threshold) in the normal state. Therefore, before the filter **11** is damaged by melting, the controller **71** can detect a malfunction in the exhaust gas aftertreatment device **10** by detecting that the index exceeds an upper limit value.

(34) Note that the index may be obtained by multiplying “a/b” by a coefficient learned from past data. For example, the index may be obtained by multiplying a numerical value of “g (a/b)” including “a/b” by a coefficient f(t) that changes with time t. Specifically, for example, “f(t)=t” and “g(a/b)=(a/b).sup.2” may be allowable, and in this case, the index increases with the lapse of time.

(35) Note that the values of “a” and “b” in the index may be weighted. For example, the value of “a” may be set to have a greater weight than the value of “b”. Further, the index may be a product “a×b” obtained from “a” and “b”. In this case, the upper limit value (threshold) of the index in the normal state is set in consideration of a fact that the value of “b” increases when the filter **11** is clogged and the value of “a” increases when the filter **11** has a high temperature. Before the filter **11** is damaged by melting, the controller **71** can detect a malfunction in the exhaust gas aftertreatment device **10** by detecting that the index exceeds the upper limit value (threshold).

(36) As described above, a malfunction in the exhaust gas aftertreatment device **10** is determined based on the filter inlet-side temperature, the filter outlet-side temperature, the differential pressure between the filter inlet-side pressure and the filter outlet-side pressure, and the intake air pressure of the engine **51**. The deposition amount of fine particles deposited on the filter **11** can be obtained from the differential pressure between the filter inlet-side pressure and the filter outlet-side pressure. A temperature state of the filter **11** is obtained from the filter inlet-side temperature and the filter outlet-side temperature. Therefore, before the filter **11** is damaged by melting, a malfunction in the exhaust gas aftertreatment device **10** can be detected by using the differential pressure between the filter inlet-side pressure and the filter outlet-side pressure, the filter inlet-side temperature, and the filter outlet-side temperature for the determination. In the present embodiment, unlike the PM sensor of Patent Literature 1, the outputs from the temperature sensors **2**, the differential pressure sensor **3**, and the intake air pressure sensor **4** do not change significantly. Therefore, the malfunction in the exhaust gas aftertreatment device **10** can be detected with high accuracy.

(37) Further, in the present embodiment, a malfunction in the exhaust gas aftertreatment device **10** is determined based on the index that is the value obtained by multiplying the ratio between the filter inlet-side temperature and the filter outlet-side temperature by a ratio between the differential pressure between the filter inlet-side pressure and the filter outlet-side pressure and the intake air pressure of the engine **51**. Since such an index is dimensionless, a malfunction in the exhaust gas aftertreatment device **10** can be detected with high accuracy by using this index for any type of the work machine **20**.

(38) The controller **71** (specifically, the notification means) may perform control for notifying a determination result in a case where the controller **71** (specifically, the determination means) determines that the exhaust gas aftertreatment device **10** has malfunctioned. Specifically, for example, in a case where the work machine **20** includes a notification device, in the work machine **20**, when the notification device notifies the determination result, a malfunction in the exhaust gas aftertreatment device **10** is notified to the operator who operates the work machine **20**. The notification device may be, for example, a speaker or a display provided in the cab **23** of the work machine **20**. In addition, the controller **71** may transmit a fact that the exhaust gas aftertreatment device **10** has malfunctioned to an external device such as a server located outside the work machine **20**, via the communication device **74**. In this case, a malfunction in the exhaust gas aftertreatment device **10** can be notified to a person outside the work machine **20**. Therefore, for example, in a case where the work machine **20** is automatically operated, the operation can be stopped by operating the work machine **20**. Further, the controller **71** may transmit the fact that the exhaust gas aftertreatment device **10** has malfunctioned to a terminal device for remotely operating the work machine **20**. In this case, the malfunction in the exhaust gas aftertreatment device **10** can be notified to a remote operator who performs a remote operation. As described above, it is possible to prevent the work machine **20** from continuing to operate in a situation where the filter **11** is likely to be damaged by melting.

(39) Further, the controller **71** (specifically, the limiting means) may perform control for limiting the output from the power source within a predetermined range in a case where the controller **71** (specifically, the determination means) determines that the exhaust gas aftertreatment device **10** has malfunctioned. Specifically, the controller **71** limits an output from at least one of the engine **51**, the hydraulic pump **52**, and the control valve group **53** within the predetermined range.

(40) When the upper limit value of the output from the power source is limited, the damage of the filter **11** due to melting can be reduced. In addition, in a case where the lower limit value of the output from the power source is limited and the load applied to the engine **51** is increased, the regeneration processing of the exhaust gas aftertreatment device **10** can be assisted. The regeneration processing is processing for increasing the load on the engine **51** and heating the filter **11** to burn the fine particles deposited on the filter **11**.

(41) Further, the controller **71** (specifically, the accumulated time calculation means) may calculate an accumulated time during which the operation device **28** outputs an operation command in a state where the controller **71** (specifically, the limiting means) limits the output from the power source. The controller **71** stores the accumulated time in the storage device **73**. As a result, a determination is made that the exhaust gas aftertreatment device **10** has malfunctioned, and it can be seen how long the movable portions have been operating in a state where the output from the power source is limited. The controller **71** then transmits the accumulated time stored in the storage device **73** from the communication device **74** to the outside, for example, at timing when the engine **51** is stopped.

(42) The controller **71** (specifically, the security processing means) performs processing relating to information security on at least one of the values detected by the temperature sensors **2**, the differential pressure sensor **3**, and the intake air pressure sensor **4**. Specifically, at least one of the temperature sensors **2**, the differential pressure sensor **3**, and the intake air pressure sensor **4** gives a signature to the detection value to be transmitted to the controller **71**. The controller **71** verifies the signature, and in a case where the verification succeeds, calculates an index based on the detection value to which the signature is given. When verifying the signature given to the detection value transmitted to the controller **71**, the controller **71** can detect modification of the detection value, and can check a transmission source to prevent spoof. Accordingly, the security of the detection value can be ensured. Note that an external device such as a server outside the work machine **20** may verify the signature. In addition, the controller **71** itself is desirably mounted in a region having tamper resistance. For example, it is desirable that the controller **71** is stored in a housing so as to be hardly removed in an unauthorized manner, or to enable detection of unauthorized removal of the controller **71** from the housing.

(43) Note that at least one of the temperature sensors **2**, the differential pressure sensor **3**, and the intake air pressure sensor **4** may encrypt the detection value to be transmitted to the controller **71**. The encryption may be public key encryption or common key encryption. The controller **71** decrypts the encrypted detection value, and calculates an index based on the decrypted detection value when the decryption succeeds. When the controller **71** decrypts the encrypted detection value transmitted to the controller **71**, high confidentiality can be maintained. Accordingly, the security of the detection value can be ensured. Note that an external device such as a server outside the work machine **20** may decrypt the detection value. Further, the controller **71** desirably stores the encrypted detection value in an area having tamper resistance (a chip or the like having tamper resistance against unauthorized access).

Operation of Diagnostic Device

(44) Next, the operation of the diagnostic device **1** will be described with reference to FIG. 5 which is a flowchart of diagnostic processing.

(45) First, the controller **71** activates the temperature sensors **2**, the differential pressure sensor **3**, and the intake air pressure sensor **4** (step S1). Next, the controller **71** calculates an index based on the detection values of the temperature sensors **2**, the differential pressure sensor **3**, and the intake air pressure sensor **4** (step S2). Specifically, the controller **71** detects the filter inlet-side

temperature and the filter outlet-side temperature based on the values detected respectively by the temperature sensors **2** (temperature detection step), detects the differential pressure between the filter inlet-side pressure and the filter outlet-side pressure based on the value detected by the differential pressure sensor **3** (differential pressure detection step), detects the intake air pressure of the engine **51** based on the value detected by the intake air pressure sensor **4** (intake air pressure detection step), and calculates an index using these detection results.

(46) Next, the controller **71** determines whether the exhaust gas aftertreatment device **10** is in a malfunction state (abnormal state) (step S3: determination step). In a case where a determination is made in step S3 as not being in the abnormal state (S3: NO), the controller **71** returns to step S2.

On the other hand, in a case where the determination is made in step S3 as being in the abnormal state (S3: YES), the controller **71** notifies the determination result (step S4). Then, the controller **71** limits the output from the power source within a predetermined range (step S5), and ends this flow.

(47) As described above, the diagnostic device **1** according to the present embodiment determines a malfunction in the exhaust gas aftertreatment device **10** based on the filter inlet-side temperature, the filter outlet-side temperature, the differential pressure between the filter inlet-side pressure and the filter outlet-side pressure, and the intake air pressure of the engine **51**. The deposition amount of fine particles deposited on the filter **11** can be obtained from the differential pressure between the filter inlet-side pressure and the filter outlet-side pressure. A temperature state of the filter **11** is obtained from the filter inlet-side temperature and the filter outlet-side temperature. Therefore, before the filter **11** is damaged by melting, a malfunction in the exhaust gas aftertreatment device **10** can be detected by using the differential pressure between the filter inlet-side pressure and the filter outlet-side pressure, the filter inlet-side temperature, and the filter outlet-side temperature for the determination. In the diagnostic device **1** according to the present embodiment, unlike the PM sensor of Patent Literature 1, the outputs from the temperature sensors **2**, the differential pressure sensor **3**, and the intake air pressure sensor **4** do not change significantly. Therefore, the malfunction in the exhaust gas aftertreatment device **10** can be detected with high accuracy.

(48) Further, in the present embodiment, the controller **71** can determine a malfunction in the exhaust gas aftertreatment device **10** based on the index that is the value obtained by multiplying the ratio between the filter inlet-side temperature and the filter outlet-side temperature by the ratio between the differential pressure between the filter inlet-side pressure and the filter outlet-side pressure and the intake air pressure of the engine **51**. Since such an index is dimensionless, a malfunction in the exhaust gas aftertreatment device **10** can be detected with high accuracy by using this index for any type of the work machine **20**.

(49) Further, in the present embodiment, in a case where a determination is made that the exhaust gas aftertreatment device **10** has malfunctioned, the determination result can be notified. For example, in a case where the determination result is notified in the work machine **20**, the malfunction in the exhaust gas aftertreatment device **10** can be notified to the operator who operates the work machine **20**. Further, in a case where the determination result is notified to the outside of the work machine **20** via the communication device **74**, the malfunction in the exhaust gas aftertreatment device **10** can be notified to a person outside the work machine **20**. Therefore, for example, in a case where the work machine **20** is automatically operated, the operation can be stopped by operating the work machine **20**. In a case where the malfunction in the exhaust gas aftertreatment device **10** is transmitted to the terminal device for remotely operating the work machine **20**, the malfunction in the exhaust gas aftertreatment device **10** can be notified to the remote operator who performs the remote operation. Therefore, it is possible to prevent the work machine **20** from continuing to operate in a situation where the filter **11** is likely to be damaged by melting.

(50) Further, in the present embodiment, in a case where a determination is made that the exhaust gas aftertreatment device **10** has malfunctioned, the output from the power source can be limited within a predetermined range. When the upper limit value of the output from the power source is

limited, the damage of the filter **11** due to melting can be reduced. In addition, in a case where the lower limit value of the output from the power source is limited and the load applied to the engine **51** is increased, the regeneration processing of the exhaust gas aftertreatment device **10** can be assisted.

(51) Further, in the present embodiment, it is possible to calculate the accumulated time during which the operation device **28** outputs an operation command in a state where the output from the power source is limited. As a result, a determination is made that the exhaust gas aftertreatment device **10** has malfunctioned, and it can be seen how long the movable portions have been operating in a state where the output from the power source is limited.

(52) Further, in the present embodiment, processing relating to information security can be performed on at least one of the values detected respectively by the temperature sensors **2**, the differential pressure sensor **3**, and the intake air pressure sensor **4**. As a result, for example, modification of a detection value is detected or the detection value is encrypted, thus securing the security of information.

(53) Although the embodiment of the present invention has been described above, only specific example has been described, and the present invention is not particularly limited to the embodiment, and a specific configuration and the like can be modified in design as appropriate. In addition, the actions and effects described in the embodiment of the present invention merely recite the most suitable actions and effects resulting from the present invention, and the actions and effects according to the present invention are not limited to those described in the embodiment of the present invention.

Claims

1. A diagnostic device for an exhaust gas aftertreatment device that has a filter for collecting fine particles from exhaust gas of an engine and is provided in a vehicle, the diagnostic device diagnosing a malfunction in the exhaust gas aftertreatment device, the diagnostic device comprising: a temperature detection device that detects a filter inlet-side temperature and a filter outlet-side temperature; a differential pressure detection device that detects a differential pressure between a filter inlet-side pressure and a filter outlet-side pressure; an intake air pressure detection device that detects an intake air pressure of the engine; and a controller that determines the malfunction in the exhaust gas aftertreatment device based on values detected by the temperature detection device, the differential pressure detection device, and the intake air pressure detection device, wherein the controller determines the malfunction in the exhaust gas aftertreatment device based on an index, and wherein the index is a value based on a ratio between the filter inlet-side temperature and the filter outlet-side temperature and a ratio between the differential pressure and the intake air pressure of the engine.
2. The diagnostic device for the exhaust gas aftertreatment device according to claim 1, wherein the controller performs control for notifying a determination result in a case where the controller determines that the exhaust gas aftertreatment device has malfunctioned.
3. The diagnostic device for the exhaust gas aftertreatment device according to claim 1, wherein the vehicle is a work machine, the work machine includes a plurality of movable portions, and a power source that supplies power to the plurality of movable portions, and the controller performs control for limiting an output from the power source within a predetermined range in a case where the controller determines that the exhaust gas aftertreatment device has malfunctioned.
4. A diagnostic device for an exhaust gas aftertreatment device that has a filter for collecting fine particles from exhaust gas of an engine and is provided in a vehicle, the diagnostic device diagnosing a malfunction in the exhaust gas aftertreatment device, the diagnostic device comprising: a temperature detection device that detects a filter inlet-side temperature and a filter outlet-side temperature; a differential pressure detection device that detects a differential pressure

between a filter inlet-side pressure and a filter outlet-side pressure; an intake air pressure detection device that detects an intake air pressure of the engine; and a controller that determines the malfunction in the exhaust gas aftertreatment device based on values detected by the temperature detection device, the differential pressure detection device, and the intake air pressure detection device, wherein the vehicle is a work machine, and the work machine includes a plurality of movable portions, and a power source that supplies power to the plurality of movable portions, and wherein the controller performs control for limiting an output from the power source within a predetermined range in a case where the controller determines that the exhaust gas aftertreatment device has malfunctioned, wherein the diagnostic device further comprises an operation device that outputs operation commands for operating the movable portions, wherein the controller calculates an accumulated time during which the operation device outputs the operation commands in a state where the output from the power source is limited.

5. The diagnostic device for the exhaust gas aftertreatment device according to claim 1, wherein the controller performs processing relating to information security on at least one of the values detected respectively by the temperature detection device, the differential pressure detection device, and the intake air pressure detection device.

6. A diagnostic method for an exhaust gas aftertreatment device that has a filter for collecting fine particles from exhaust gas of an engine and is provided in a vehicle, the diagnostic method for diagnosing a malfunction in the exhaust gas aftertreatment device, the diagnostic method comprising: a temperature detection step of detecting a filter inlet-side temperature and a filter outlet-side temperature; a differential pressure detection step of detecting a differential pressure between a filter inlet-side pressure and a filter outlet-side pressure; an intake air pressure detection step of detecting an intake air pressure of the engine; and a determination step, which is executed by a controller, of determining the malfunction in the exhaust gas aftertreatment device based on values detected in the temperature detection step, the differential pressure detection step, and the intake air pressure detection step, wherein the controller determines the malfunction in the exhaust gas aftertreatment device based on an index, and wherein the index is a value based on a ratio between the filter inlet-side temperature and the filter outlet-side temperature and a ratio between the differential pressure and the intake air pressure of the engine.

7. A diagnostic program for an exhaust gas aftertreatment device that has a filter for collecting fine particles from exhaust gas of an engine and is provided in a vehicle, the diagnostic program for causing a computer to diagnose a malfunction in the exhaust gas aftertreatment device, the diagnostic program comprising causing the computer to determine the malfunction in the exhaust gas aftertreatment device based on a filter inlet-side temperature, a filter outlet-side temperature, a differential pressure between a filter inlet-side pressure and a filter outlet-side pressure, and an intake air pressure of the engine, wherein the computer determines the malfunction in the exhaust gas aftertreatment device based on an index, and wherein the index is a value based on a ratio between the filter inlet-side temperature and the filter outlet-side temperature and a ratio between the differential pressure and the intake air pressure of the engine.
